

- 2 A butterfly is a flying insect. One species of butterfly, the cabbage white butterfly, *Pieris brassicae*, is shown in Figure 2.1.



Figure 2.1

Some scientists are carrying out a long-term project to estimate the population size of all butterfly species in the UK. The scientists have asked people living in the UK to help them to collect the data.

A modified belt transect is one of the methods used to collect the data.

The recommended method for the modified belt transect is to:

- walk at a slow pace along a transect line
- identify all the butterfly species within 2.5m on each side of the transect line
- count the number of individuals of each species within 2.5m on each side of the transect line.

- (a) (i) Suggest how each species of butterfly could be identified.

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..... [1]

- (ii) The scientists standardised the width of the belt transect.

State **two other** variables that should be standardised when carrying out this belt transect.

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..... [2]



(iii) Discuss the limitations of the method used to collect data.

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..... [3]

(b) The scientists plan to compare the diversity of butterfly species found in urban areas (towns and cities) and rural areas (e.g. farmland and woodland).

Predict whether a greater diversity of butterfly species is found in urban areas **or** rural areas. Explain your answer.

prediction

explanation

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..... [2]

- (c) In another study, the number of individuals of the cabbage white butterfly, *Pieris brassicae* (shown in Figure 2.1), was counted each day over a four-year period and recorded. For each year, the highest number of butterflies counted in one day was identified.

The results are shown in Figure 2.2.

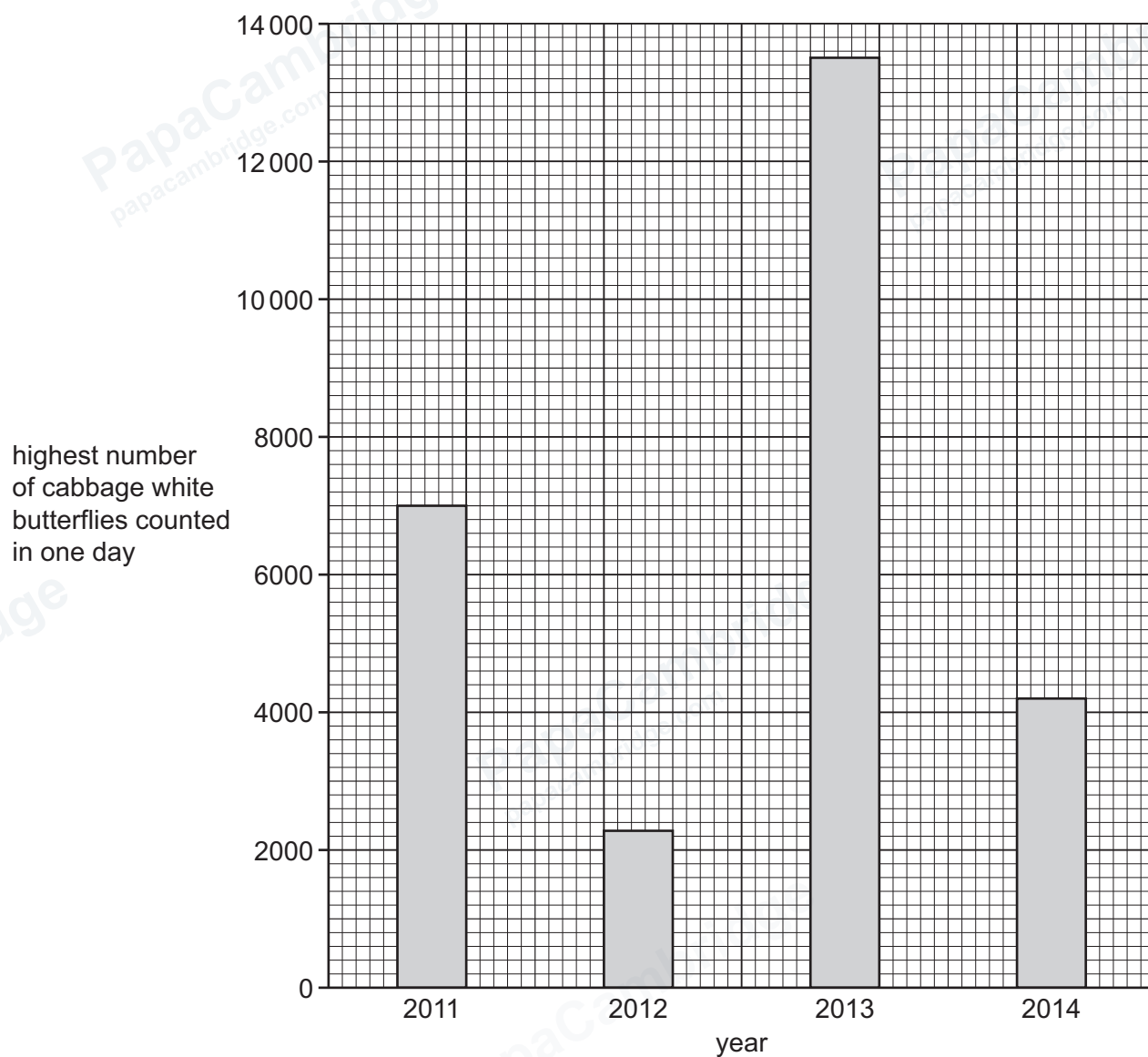


Figure 2.2



Figure 2.2 shows that there is an increase in the highest number of cabbage white butterflies counted in one day from 2012 to 2013.

Use Figure 2.2 to calculate the percentage increase from 2012 to 2013.

Give your answer to **three** significant figures.

Show your working.

percentage increase = % [2]

[Total: 10]